Joule

Dynamic Power Envelope Management for Compute Clusters

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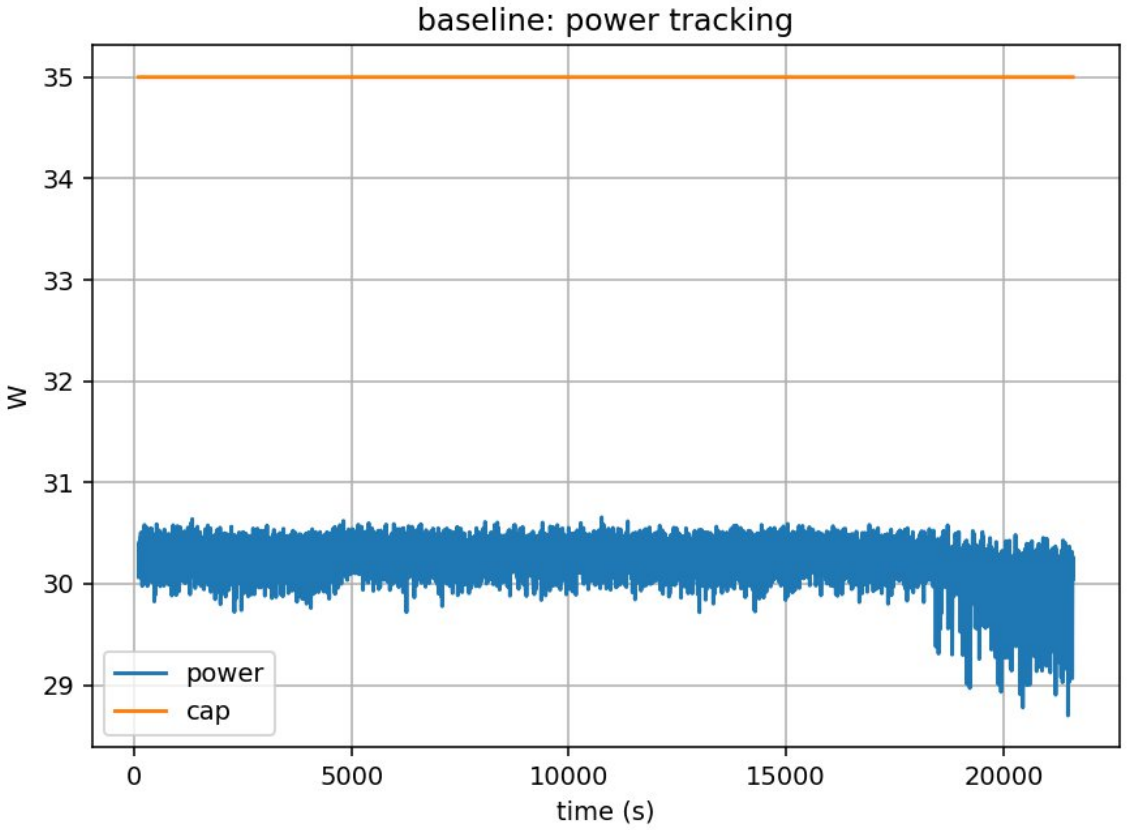
Static Power Limits Waste Compute Capacity



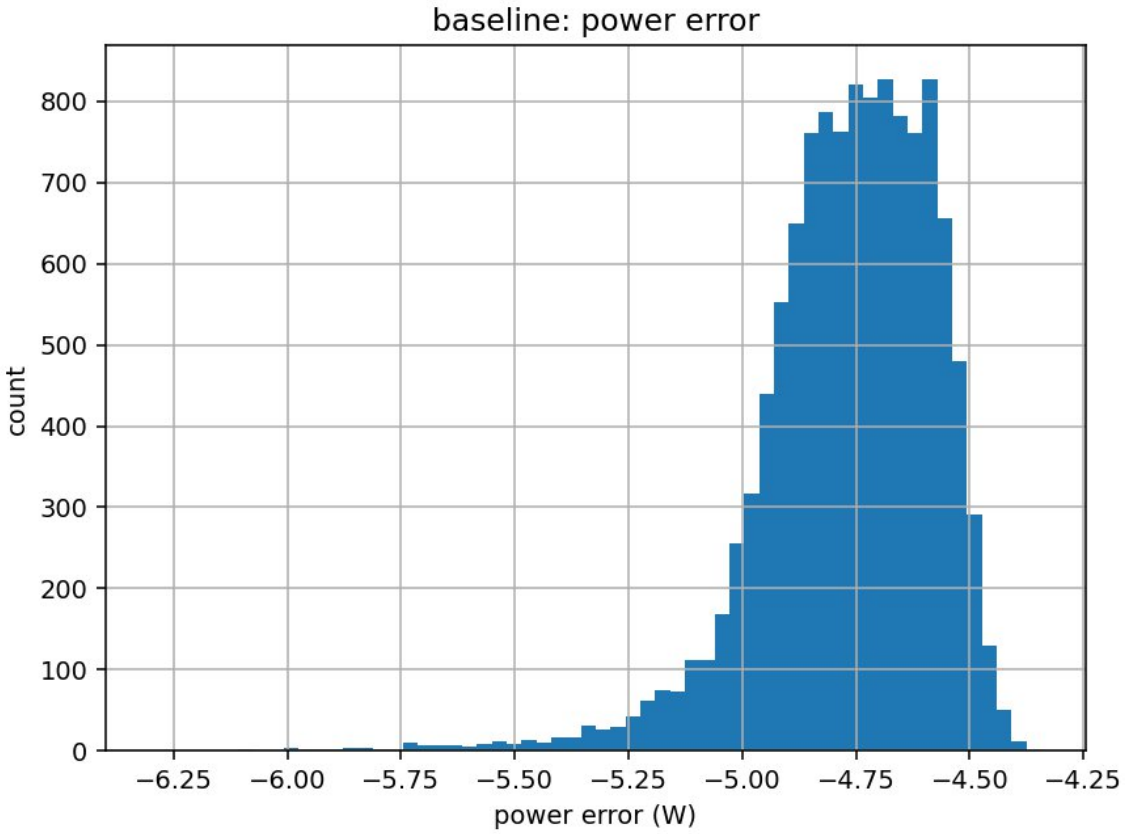
- Real workloads fluctuate significantly
- Static limits leave ~10-20% power unused

10% unused power => ~10% lost compute

Unused Power Limits Compute Throughput

- Real workloads fluctuate significantly
- Static limits leave 10-20% of power unused
- Power-constrained clusters lose potential compute

10% unused power = huge lost capacity



Joule Dynamically Allocates Power

```
+---------------+
| Joule Brain |
+---------------+
/                 \
Node Agent A   Node Agent B
|                 |
Power            Power
Sensors          Sensors
|                 |
Hardware Power Interfaces
```

- Treats power as a shared cluster resource
- Continuously adjusts node power limits
- Maintains strict compliance with a global power cap

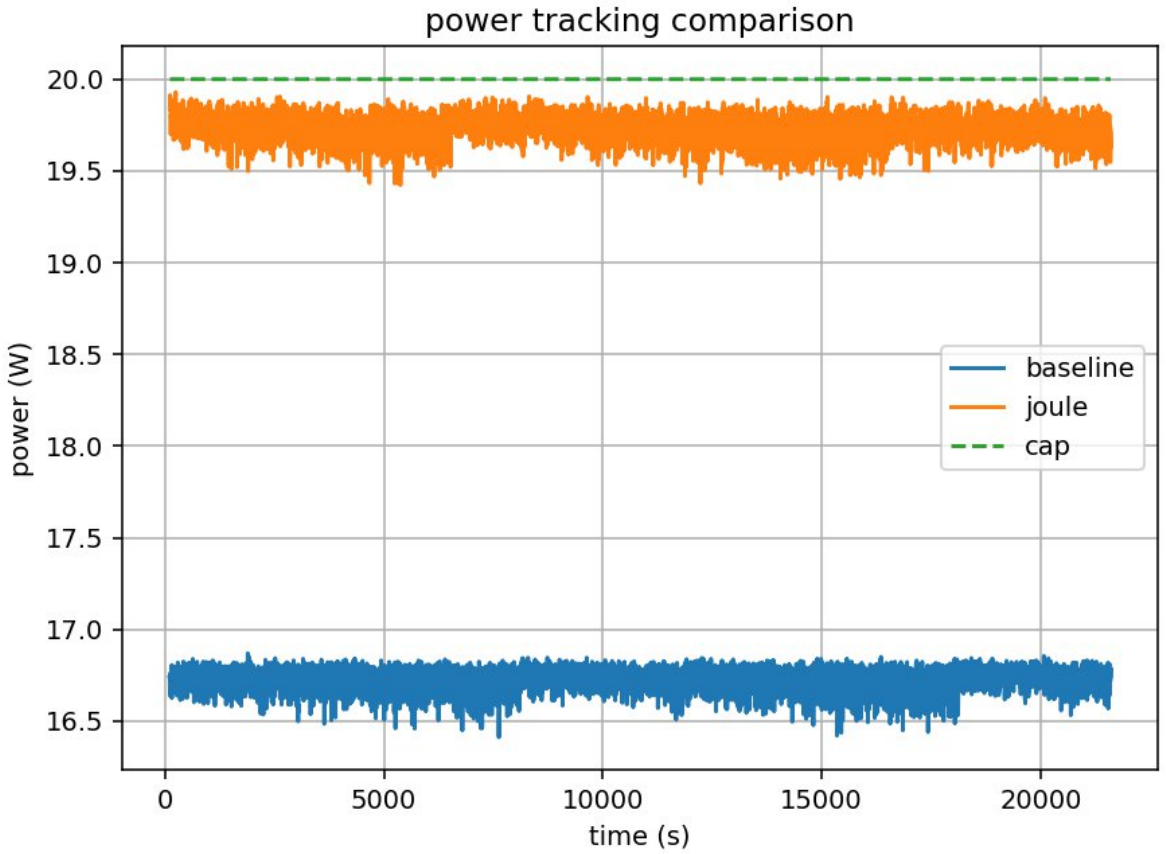
Controlled Evaluation

- Multi-node compute cluster
- Mixed compute workloads
- Multiple power-cap levels
- Six-hour runs per configuration

Same Hardware
Same Workloads
Same Power Cap
|

| |
Baseline Run Joule Run
Static Power Caps Dynamic Control
| |
---------- Compare ----------
Utilisation & Throughput

Joule Tracks the Power Cap with Near-Perfect Precision



Joule dynamically adjusts node power limits to keep cluster power near the configured envelope.

- Baseline leaves significant power headroom
- Joule tracks the envelope with <1W error
- Zero cap violations

Dynamic Power Allocation Increases Throughput

Cap	Baseline Utilisation (%)	Joule Utilisation (%)
20	83	99
25	79	99
30	79	98
35	86	97

- Dynamic power control converts unused headroom into 5–12% more compute.

Power Is the Limiting Resource

Power Is the Primary Constraint

Datacentres increasingly limited by electrical capacity

Example:
100 MW datacentre
10% utilization improvement
= 10 MW additional compute capacity

Joule increases usable compute under fixed power limits

Roadmap

Current:

Prototype validation (v0.96)

Next:

Cluster-scale deployment

GPU power interfaces

Scheduler integration

Future:

Rack-level envelopes

Datacentre-scale power management

- Expand node scaling
- GPU power interfaces
- Integration with cluster scheduleres

Next Step

Deploy Joule on GPU clusters

Seeking:

- Funding for cluster-scale evaluation
- Industry partners for pilot deployments